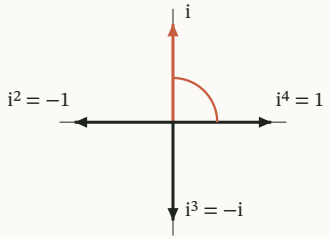
# Complex numbers CHEAT SHEET



## SQUARE ROOTS OF −1




**× i: a quarter turn**
× (−1): a half turn

$i^2 = -1$ $\quad i = \sqrt{-1}$

powers repeat every 4: $i^{35} = i^3 = -i$ $\quad i^{100} = 1$

$\frac{1}{i} = -i$ $\quad i + i^2 + i^3 + i^4 = 0$

$\sqrt{-9} = 3i$ $\quad x^2 = -16 \Rightarrow x = \pm 4i$

but $\sqrt{-4} \cdot \sqrt{-9} = 2i \cdot 3i = -6$ (not $\sqrt{36}$)

## THE ALGEBRA

$z = a + bi$: $\operatorname{Re} z = a$, $\operatorname{Im} z = b$

$a + bi = c + di \iff a = c$ and $b = d$

$(3 + 2i) + (1 + 4i) = 4 + 6i$

$(a + bi)(c + di) = (ac - bd) + (ad + bc)i$

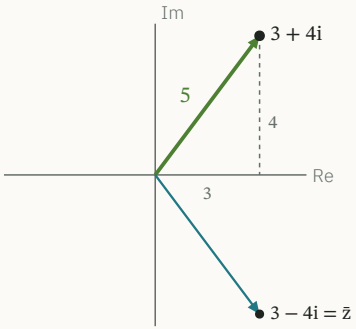
$(2 + 3i)(4 + 5i) = -7 + 22i$ $\quad (1 + i)^2 = 2i$

conjugate $\bar{z} = a - bi$ $\quad z\bar{z} = a^2 + b^2$

$\frac{3+4i}{1+2i} = \frac{11-2i}{5}$ (top and bottom $\times\, 1 - 2i$)

$\frac{1}{z} = \frac{\bar{z}}{a^2+b^2}$ $\quad \frac{1}{1+i} = \frac{1-i}{2}$

## THE ARGAND PLANE




a point, or an arrow from 0
real part across, imaginary up
add: nose to tail
$\bar{z}$: mirror in the real axis
$-z$: a half turn

## THE MODULUS

$|z| = \sqrt{(a^2 + b^2)}$ $\quad |3 + 4i| = 5$ $\quad |-5 + 12i| = 13$

$z\,\bar{z} = |z|^2$ $\quad |\bar{z}| = |z|$ $\quad |-z| = |z|$

$|z_1 z_2| = |z_1|\,|z_2|$ $\quad |z_1/z_2| = |z_1| \,/\, |z_2|$

$|z_1 + z_2| \le |z_1| + |z_2|$ (triangle inequality)

$|z - w|$ = the distance between z and w

## CONJUGATE RULES

$\overline{(z_1 + z_2)} = \bar{z}_1 + \bar{z}_2$ $\quad \overline{(z_1 z_2)} = \bar{z}_1\,\bar{z}_2$

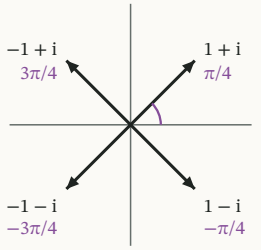
$z + \bar{z} = 2 \operatorname{Re} z$ $\quad z - \bar{z} = 2i \operatorname{Im} z$ $\quad z$ real $\iff z = \bar{z}$

## LOCI

| | |
|---|---|
| $\lvert z - a\rvert = r$ | circle, centre a, radius r |
| $\lvert z - a\rvert \le r$ | the disc (< : edge left out) |
| $\lvert z - a\rvert = \lvert z - b\rvert$ | perpendicular bisector of a, b |
| $\arg(z - a) = \theta$ | half-line from a (a left out) |

nearest point of $|z - 2 - 2i| = 2$ to 0: $2\sqrt{2} - 2 \approx 0.83$

## THE ARGUMENT




arg z: the angle from the positive real axis, in radians
$\tan\theta = b/a$ gives the same for $1 + i$ and $-1 - i$:
look at the quadrant

## POLAR FORM

$z = r(\cos\theta + i\sin\theta)$ $\quad r = |z|$, $\theta = \arg z$

$a = r\cos\theta$ $\quad b = r\sin\theta$ $\quad \tan\theta = b/a$ (check the quadrant)

principal argument $-\pi < \theta \le \pi$

$1 + i\sqrt{3} = 2(\cos\pi/3 + i\sin\pi/3)$

$-2 - 2i = 2\sqrt{2}(\cos(-3\pi/4) + i\sin(-3\pi/4))$

$4(\cos 5\pi/6 + i\sin 5\pi/6) = -2\sqrt{3} + 2i$

| θ | | cos θ | sin θ |
|---|---|---|---|
| π/6 | 30° | √3/2 | 1/2 |
| π/4 | 45° | √2/2 | √2/2 |
| π/3 | 60° | 1/2 | √3/2 |
| π/2 | 90° | 0 | 1 |

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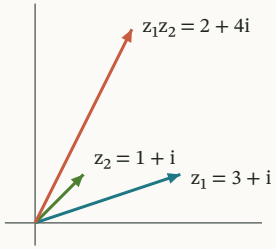
## MULTIPLY AND DIVIDE: TURN AND STRETCH




multiply: lengths multiply, angles add

$\sqrt{10} \times \sqrt{2} = \sqrt{20}$

$18.4° + 45° = 63.4°$

$z_1z_2 = r_1r_2[\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)]$

$\frac{z_1}{z_2} = \frac{r_1}{r_2}[\cos(\theta_1 - \theta_2) + i \sin(\theta_1 - \theta_2)]$

$\frac{1}{z}$: length $\frac{1}{r}$, angle $-\theta$

$\times$ i: a quarter turn $(3 + 2i)\, i = -2 + 3i$

turn by $\alpha$: $\times (\cos\alpha + i \sin\alpha)$

$2(\cos 50° + i \sin 50°) \cdot 3(\cos 70° + i \sin 70°)$

$= 6(\cos 120° + i \sin 120°) = -3 + 3\sqrt{3}\, i$

## DE MOIVRE'S THEOREM

$(\cos\theta + i \sin\theta)^n = \cos n\theta + i \sin n\theta$

$(1 + i)^8 = 16 \quad (\sqrt{3} + i)^6 = -64$

$(1 - i)^{10} = -32i \quad (1 + i)^{-2} = -i/2$

$\cos 2\theta = \cos^2\theta - \sin^2\theta \quad \sin 2\theta = 2 \sin\theta \cos\theta$

$\cos 3\theta = 4\cos^3\theta - 3\cos\theta \quad \sin 3\theta = 3\sin\theta - 4\sin^3\theta$

## ROOTS OF COMPLEX NUMBERS

$\sqrt{(3 + 4i)}$: $(x + iy)^2 = 3 + 4i$

$x^2 - y^2 = 3,\ 2xy = 4,\ x^2 + y^2 = |3 + 4i| = 5$

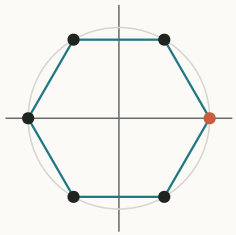
$\Rightarrow \pm(2 + i) \qquad \sqrt{(-5 + 12i)} = \pm(2 + 3i)$

n-th roots of $r(\cos\theta + i \sin\theta)$:

$r^{1/n}[\cos\frac{\theta + 2\pi k}{n} + i \sin\frac{\theta + 2\pi k}{n}],\ k = 0 \ldots n - 1$

$z^3 = 8$: $2,\ -1 \pm i\sqrt{3} \qquad \sqrt{i} = \pm(1 + i)/\sqrt{2}$

## ROOTS OF UNITY



$z^6 = 1$: a regular hexagon the roots share the turn

$\omega = -\frac{1}{2} + (\sqrt{3}/2)i = \cos 120° + i \sin 120°$

$\omega^3 = 1 \quad 1 + \omega + \omega^2 = 0 \quad \omega^2 = \bar{\omega}$

$z^3 - 1 = (z - 1)(z - \omega)(z - \omega^2)$

$z^n = 1$: $z = \cos\frac{2\pi k}{n} + i \sin\frac{2\pi k}{n}$

the n-th roots of unity add to 0

## QUADRATICS WITH COMPLEX ROOTS

$az^2 + bz + c = 0,\ D = b^2 - 4ac < 0$:

$z = \frac{-b \pm i\sqrt{(-D)}}{2a}$

$z^2 - 4z + 13 = 0$: $D = -36,\ z = 2 \pm 3i$

$z^2 + z + 1 = 0$: $z = \omega, \omega^2$

real coefficients: roots in conjugate pairs

$z^3 - z^2 + 2 = (z^2 - 2z + 2)(z + 1)$: $1 \pm i, -1$

degree n: n roots, counting repeats

## EULER'S FORMULA

$e^{i\theta} = \cos\theta + i \sin\theta \quad |e^{i\theta}| = 1$

$z = re^{i\theta} \quad 1 + i = \sqrt{2}\, e^{i\pi/4}$

$e^{i\pi} + 1 = 0 \quad e^{i\pi/2} = i \quad e^{2\pi i} = 1$

$r_1e^{i\alpha} \cdot r_2e^{i\beta} = r_1r_2e^{i(\alpha + \beta)} \quad (e^{i\theta})^n = e^{in\theta}$

$\cos\theta = \frac{e^{(i\theta)} + e^{(-i\theta)}}{2} \quad \sin\theta = \frac{e^{(i\theta)} - e^{(-i\theta)}}{2i}$

$i^i = e^{-\pi/2} \approx 0.2079 \quad$ roots of unity $e^{2\pi ik/n}$